



DEALER BEST PRACTICE SERIES

Electric Pallet Trucks

Reduce Material Handling Costs

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

Electric Pallet Trucks Reduce Material Handling Costs

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1.0 Introduction

Many dealers still transport parts and materials delivered to the Component Rebuild Centers (CRC) manually using hand pallet trucks. Some dealers have replaced the manual pallet handlers with electric pallet trucks.

This Best Practice is an economic comparison of “manual” versus “electric” material handling equipment. This Best Practice comes from Ferreyros – Lima, Peru.



2.0 Best Practice Description

Tests were conducted over a controlled period to measure the speed of delivery of parts baskets to the CRC (average distance 50 Meters) using electric pallet trucks versus manual pallet handlers. The results showed that the electric pallet trucks were 2.3 times faster than manual transport.

There were other advantages including:

- Less parts handling staff fatigue.
- Reduced / eliminated back pain incidents.
- Quick delivery means fewer parts need to be staged in the bay – delivered when needed (just in time).
- Reduced technician “wait times”.
- Handlers were small and maneuverable – able to work in narrow spaces.

Investment analysis was based on “time recovered” in transporting materials from the CRC parts receiving area to the work bays and back to receiving.

Dealer realized a 44% NPV return on investment and cost recovery period of 2 years 11 months.

3.0 Implementation Steps

The key elements of the analysis includes quantifying the “time saved” versus the pallet truck purchase and operating / maintenance cost of the equipment.

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DATE
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Time savings:

Moving Way		Speed
Electric Transpallet Speed (m/min)		56.3
Manual Transpallet Speed (m/min)		23.92
Average Distance used in Simulation (m)		50

The electric pallet trucks were 2.35 times faster than the manual pallet handlers. This information can be converted to a “man hours” saved benefit to be used in the investment analysis.

Savings are then compared to purchase, owning and operating costs, taxes, depreciation, etc. to determine a return on investment.

To determine the number of electric pallet trucks required to handle the workload, the average daily number of parts baskets moved to the work bays was examined.

Before:

(2 Mechanical Transpallets)		12 Hr Day	
Incoming Parts Baskets		714	
Delivered		372	
Backlog – Waiting to Deliver 48%		342	

After:

(2 Electrical Transpallets)		12 Hr Day	
Incoming Parts Baskets		714	
Delivered		714	
Backlog – Waiting to Deliver		0	



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There are many types of electric pallet trucks available. Models include riding trucks or walk behind trucks. There are also models with a forklift for handling pallets staged in vertical racks or for stacking.

Caterpillar offers several electric pallet trucks such as the NPV20 / 25.

4.0 Benefits

- Reduced time required to deliver parts to work bays.
- Reduced technician “waiting time” for parts.
- Reduced use of bay space to stage parts – just in time delivery process.
- Reduced fatigue and back pain for parts handling staff.
- Possible reduced manpower required.

5.0 Resources Required

- Purchase of equipment
- Maintenance and repairs of equipment.
- Operational, Safety, and Maintenance Training

6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

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